Area Binning

Binning option enables you to divide sections in designated area, called bin, and extract representative values in each bin. This option also compares and analyzes the values in each bin.

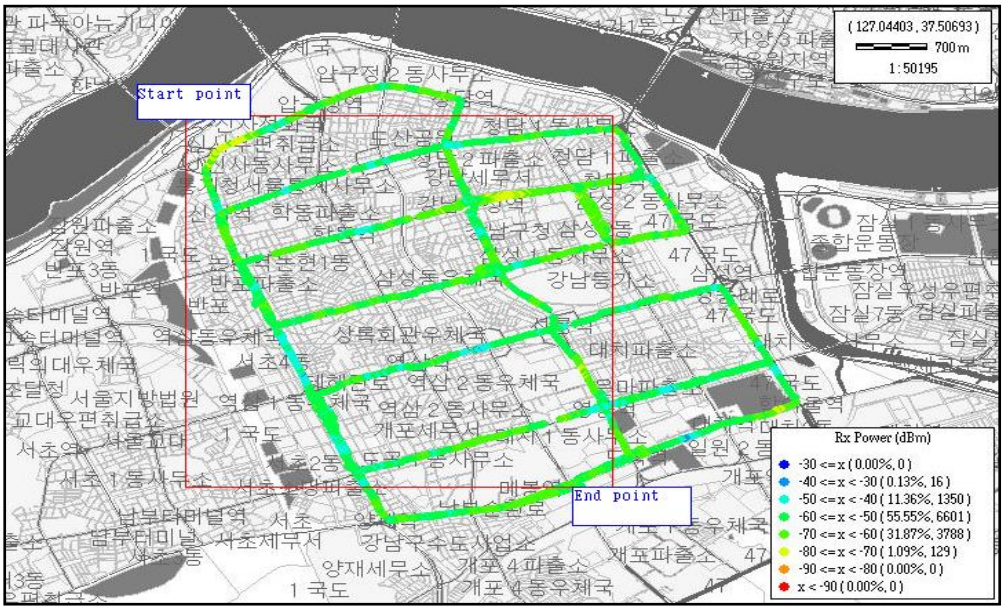
The details of Binning menu are as follows.

Item	Description
Create Area	Create user defined region.
Create Route Based Area	Create area automatically by calculating border in all directions based on measured total route.
Manage Area	Manage the created Area (delete, edit, import).
Run	Start Area Binning and Route Binning

To start binning:

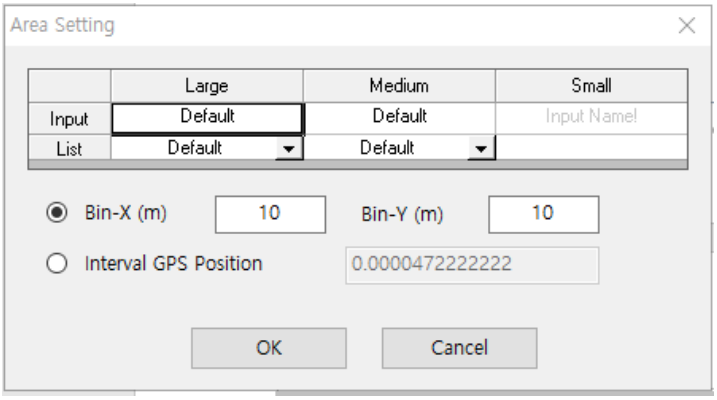
1. Right click on a map analysis window, and go to Binning – Create Area.
2. Select area you want to create bin by dragging on the map analysis window.

   You may change area selection mode from Map mouse menu – Data Selection – Rectangle/Polygon/Circle Mode.



3. Area Setting window appears.

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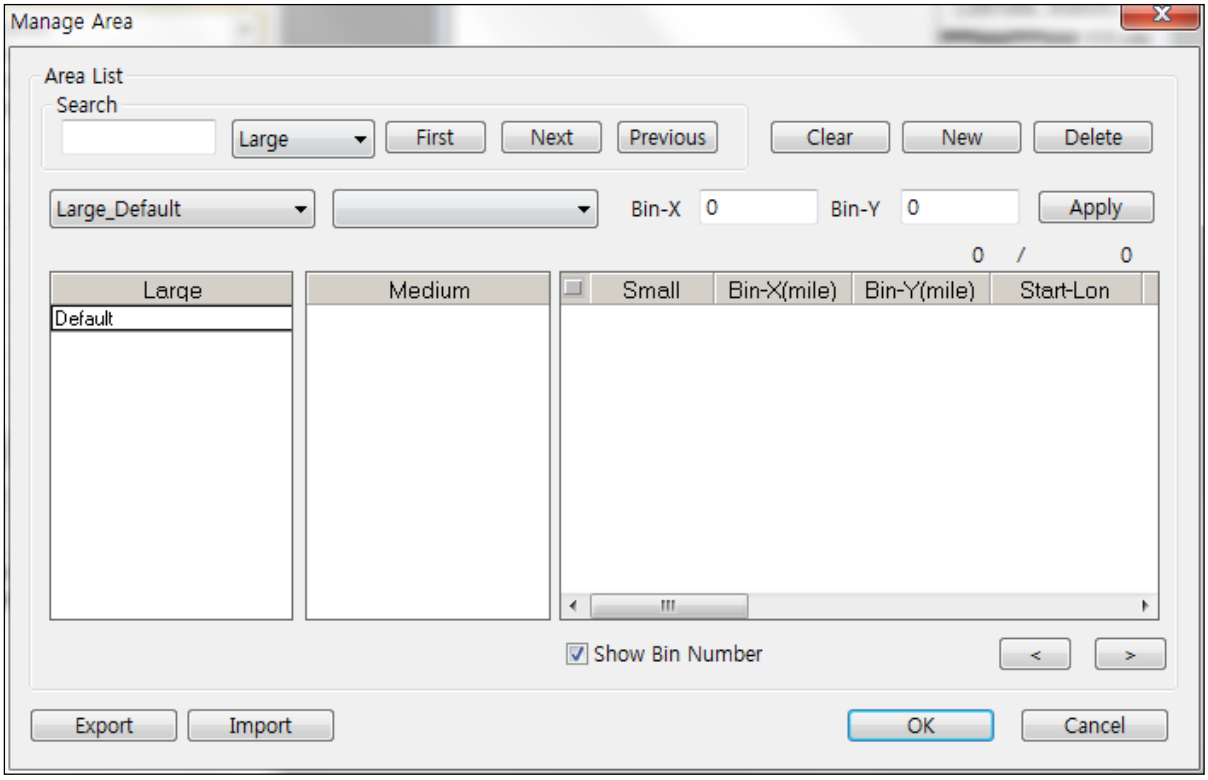


4. Set area and bin size.

Item	Description
Interval GPS Position	User can set the interval of GPS.
Input	Double click on Input cell and enter Large, Medium, Small area names. Large, Medium, Small area are designed to manage multiple areas based on the size and relationship of area. Large area contains Medium areas, and Medium area contains Small areas. You may find Large, Medium, Small area names from Manage Area window.
List	Apply existing name.
Bin-X, Bin-Y	Enter Bin X and Y axis size.

5. Click OK.
6. Right click on a map analysis window, and go to Binning – Manage Area.
7. Manage Area window appears.

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8. Set area binning options.

The detailed description of each option is as follows.

Item	Description
Search	Search a certain area (Large/Medium/Small).
Clear	Initiate all option settings in Area Management window.
New	Create a new area. You may change longitude and latitude of starting and ending point by double-clicking directly on Start-Lon/Lat, End-Lon/Lat cells and enter value you want in the entry field.
Delete	Delete the current area.
Display	Select checkboxes in front of area name to determine if show or hide the area in the map analysis window.
Large	Show name of large area. Large, Medium, Small area levels show that the final Small area is included exactly in which Large and Medium area. Default name: Default
Medium	Show name of medium area. Large, Medium, Small area levels show that the final Small area is included exactly in which Large and Medium area. Default name: Default
Small	Show name of small area. Large, Medium, Small area levels show that the final Small area is included exactly in which Large and Medium area. Default name: New SArea# (#: line number)
Bin-X, Bin-Y	Enter bin X and Y axis size in the entry field and click Apply button.
Export	Export the current option settings in excel.
Import	Import existing option settings in excel.
Show Bin Number	If you check this option, the numbers are not displayed in the squares (Bin).
Prev	Go to previous page.
Next	Go to next page.
OK	Save the current option settings and close the window.
Cancel	Cancel the current option settings and close the window.

9. Click OK.

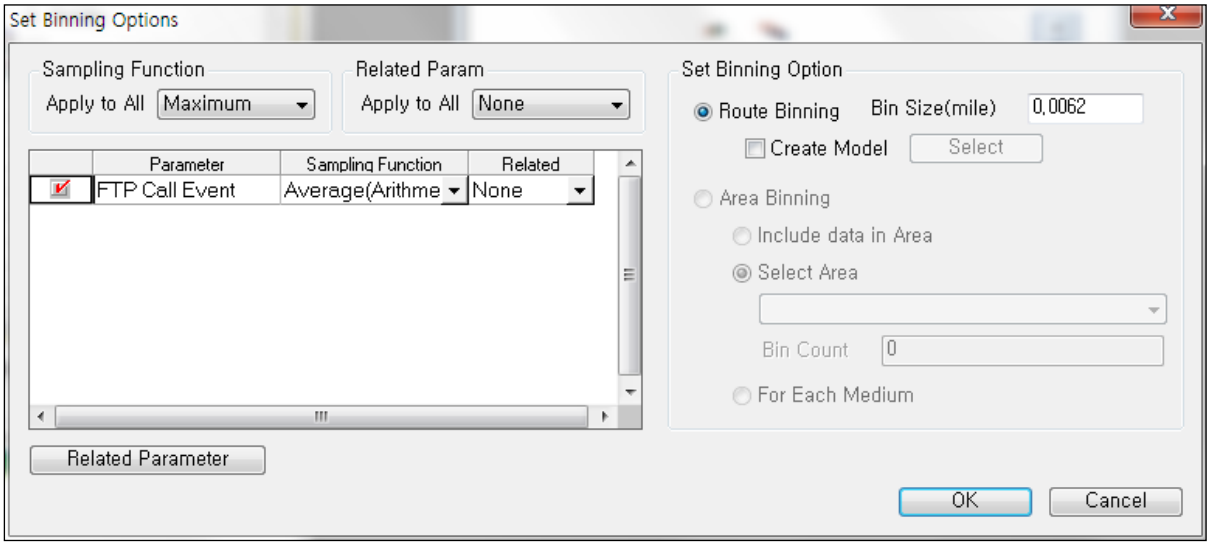
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10. Grid is displayed over the selected area on the map.

    If you enlarge the map analysis window by using Zoom In option, Bin numbers are displayed in each bin.



11. Right click on the map analysis window, and go to Binning – Run. Set Binning Options window appears.



12. Set options for binning.

The detailed description of each option is as follows.

Item	Description
Sampling Function	Select formula for representative value to be displayed in each bin by clicking drop-down button. ◆ Apply to All: Apply formula for all parameters selected. This option is useful when more than one parameter are opened in a map analysis window. ◆ Average: Display average value of each bin at the center position based on time. ◆ Medium: Display nearest value of average in each bin at its original

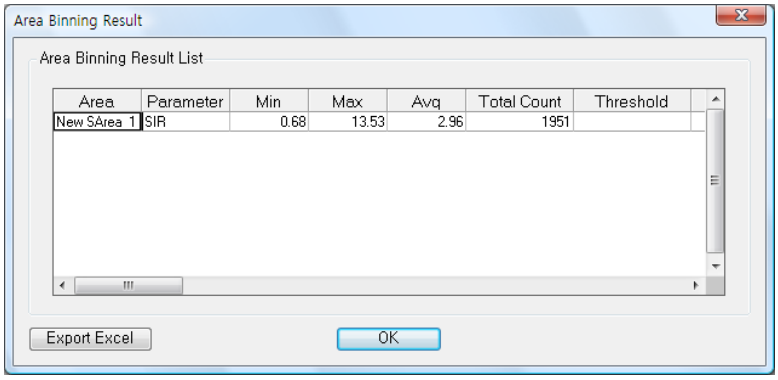
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Item	Description
	position. If there are identical values at more than one position, Medium value is displayed at the firstly founded position. ◆ Median: Display middle size data at its original position. If there are identical values at more than one position, middle size data is displayed at the firstly founded position. ◆ Maximum: Display maximum value of each bin at its original position. If there are identical values at more than one position, max value is displayed at the firstly founded position. ◆ Minimum: Display minimum value of each bin at its original position. If there are identical values at more than one position, min value is displayed at the firstly founded position. ◆ First: Display the first data based on time in each bin at its original position. ◆ Last: Display the last data based on time in each in at its original position. ◆ Mode: Display mode value at the center in time order. If you run binning for Sector and PSC Type parameters, only Mode value is displayed whichever formula you select.
Related Param	Select related parameter. Click Related Parameter button at the left bottom of the window to set related parameter. ◆ Apply to All: Apply related parameter for all parameters selected. This option is useful when more than one parameters are opened in a map analysis window. Note) Related Param function performs only when Area Binning is checked.
Set Binning Option	◆ Route Binning: Perform binning based on specified bin size along the route. ◆ Bin Size (m): Enter bin size for route binning in the entry field. ◆ Create Model: Create a new model with data derived from Route Binning. Click Select button to select parameters. ◆ Area Binning: Perform binning based on the created area. ◆ Include data in Area: Choose bins including data and perform binning. ◆ Select Area: Select area. You may select among all area set from Manage Area window. ◆ Bin Count: Show number of bins in the created area. ◆ For Each Medium: Perform binning per Medium area.
Related Parameter	Display the related parameter value occurred the same time with Binning result (possible for all function except Average(Linear/arithmetic)). The related parameter should be configured by user.
OK	Save setting and start binning.
Cancel	Cancel setting and close Set Binning Options window.

13. Click OK.

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14. Area Binning Result window appears.

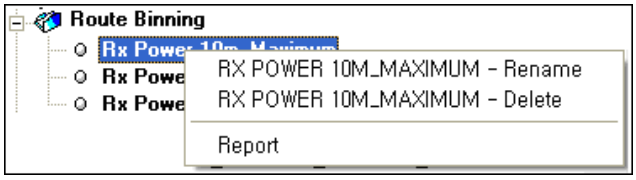


The detailed description of each item in Area Binning Result window is as follows.

Item	Description
Area	Show name of binned area.
Parameter	Show selected parameter.
Min	Show minimum value among representative values for each bin.
Max	Show maximum value among representative values for each bin.
Avq	Show average value among representative values for each bin.
Total Count	Show total number of representative values.
Threshold	Enter threshold value by double-clicking on the entry field.
More Than Threshold	Show number of values over the threshold entered in Threshold cell.
Less Than Threshold	Show number of values under the threshold entered in Threshold cell.
More Than Threshold Ratio	More Than Threshold/Total Count * 100
Less Than Threshold Ratio	Less Than Threshold/Total Count * 100
Export Excel	Export the binning result in xls.

15. Binned data can be found from Route Binning and Area Binning from parameter tree of Graph, Map, Table, CDF, and PDF analysis tab.

You may rename and delete the binned parameters under Route Binning and Area Binning tree by right clicking the parameter and select on Rename and Delete.



16. Rename and Delete binning parameter.

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